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New product development team intelligence: Antecedents and consequences

Ali E. Akgün [a,*], Mumin Dayan [b], Anthony Di Benedetto [c]

[a] Science and Technology Studies, Gebze Institute of Technology, Turkey
[b] School of Business Administration, University of Sharjah, United Arab Emirates
[c] School of Business Adminstration, Temple University, USA

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ABSTRACT

Our study investigated the effect of team knowledge on new product development (NPD). By investigating 207 NPD projects, we found that the declarative and procedural knowledge of the team and their use of IT had a positive influence on the team's knowledge base; and that the higher the functional diversity of the project team, the greater their overall knowledge. We also found that team knowledge positively impacted new product creativity and success in the market place.

1. Introduction

The importance of new product development (NPD) teams has been emphasized in the last decade in the technology and innovation management literature [19]. Many papers have pointed out that most successful NPD projects were achieved through the collective efforts of individuals on the teams [1] and have suggested approaches for managers to form and manage NPD teams; cross-functional integration [13], team learning [15], knowledge management (KM), and collaborative technologies [31] are some of the ways. One factor discussed recently has been *team intelligence* [4]; it seemed important because it helped the team to promote effective knowledge creation, fuel the learning process, and develop an effective way to implement the product. However, NPD team intelligence has been narrowly defined and has received limited empirical attention. The current *conceptual* definition of NPD team intelligence was restricted to the cognitive view of intelligence omitting its behavioral dimension. In organizational behavior literature, an entity demonstrates knowledge when it responds to changing conditions, problems, etc. in a goal-directed adaptive manner by modifying its behavior [11]. Intelligence involves adaptive behavior or *responsiveness*. Also related to the conceptual issues, the NPD team intelligence construct has been operationalized narrowly as the ability of the team to acquire, disseminate and implement/use information. However, team intelligence is a multi-dimensional construct involving a variety of capabilities and should be operationalized as a higher-order, multifaceted construct embodying both information processing and responsiveness capabilities to capture the complex nature of the process of product development. Akgün et al. [5] argued that, while information processing capabilities highlighted the internal structures and processes, it put the environment in a passive role. On the other hand, they noted that responsiveness capabilities suggested a potential reaction and sensitivity to the external environment though it omitted information processing capabilities. Thus, amalgamating the information processing and responsiveness perspectives as a higher-order construct could result in a more comprehensive view of NPD team intelligence. Further, from a managerial point of view, the antecedents and consequences of NPD team intelligence, such as factors that impact NPD team intelligence and how this influences the project outcomes, should be investigated empirically. Investigating the determinants of team intelligence could aid project managers in understanding how to elevate the team's capabilities, and how to leverage those capabilities to result a successful NPD project.

Thus our study elaborated on research on NPD team intelligence and its antecedents and consequences to provide a KM view

* Corresponding author. Fax: +90 262 605 1405.
E-mail address: aakgun@gyte.edu.tr (A.E. Akgün).

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of NPD teams and project management. In particular, we considered how cross-functional NPD project teams enhanced their information processing and responsiveness capabilities and the impact of them on NPD project outcomes.

2. Background

2.1. Team intelligence and NPD

The term *intelligence* is used in many areas and disciplines such as, individual and experimental psychology [29], group behavior [33], organizational theory, KM, organizational learning [3], and sociology [14]. Discussion of team intelligence has considered it to be a subset or subprocess of KM. Nonaka and Takeuchi [23] said that NPD teams demonstrate their intelligence by acquiring, interpreting, disseminating and using information and knowledge during the project. Chou et al. [9] stated that information-processing capabilities were the captured organizational intelligence. Thus KM overlaps with the team intelligence. However, studies on team intelligence improved the concept of KM by highlighting the value of information processing *ability* in a project. Specifically, NPD team intelligence denoted a team's capability to use information processes through project related activities that achieved a desired end or performed a particular function or value activity during the project. In the context of a KM process in an NPD project [22]; *information acquisition ability* was referred to the team's capability to collect primary or secondary information from customers, competitors, and other third parties. *Information dissemination ability* was the team's capacity to diffuse and transmit the information among relevant members of the team, involving formal and informal information transmission via interpersonal interactions, meetings, memos, etc. *Information utilization ability* was referred to the project team's ability to use of the information indirectly in strategy-related actions and to direct applications of information to influence marketing strategy-related actions. Such actions included giving meaning to data, interpreting and categorizing it, and applying the information/knowledge to solve product related problems during the NPD process. However, the information-processing capability/ability view of intelligence, which fits the cognitive scheme of the team KM process, explained one side of the NPD team intelligence. It was not sufficient to address issues that only related to the collective nature of work in project teams. In general, the behavioral perspective of intelligence shows the responsiveness of an entity. In an NPD project team, we contend that *responsiveness* refers to the actions taken in response to information and knowledge generated, transmitted, and utilized. Without responding to the environment, market information processes would provide little or no help to teams in achieving their goals.

NPD team intelligence is imperative for the effective operation and performance of the team. However, for a systematic understanding of NPD team intelligence, the antecedent factors should also be investigated to understand how we can increase the team intelligence. Consistent with the study by Moorman and Miner [20], we focused on performance measures by selecting routinely studied items: new product creativity (NPC) and new product success (NPS); these are critical indicators for managers. In addition, the knowledge base (declarative and procedural knowledge), information technology usage, and cross-functional diversity have been identified as determinants of NPD team intelligence. Although these antecedents have been examined in the KM and organizational intelligence literature, there has been no empirical evidence to show how they influence NPD team intelligence.

3. Hypothesis development

3.1. Antecedents

The knowledge base is the basis of organizational intelligence [18]. In her study on organizational intelligence, Glynn [12] also emphasized its importance. In a similar fashion, NPD studies have emphasized its effect on the team's information processing and response ability. Madhavan and Grover [17] stated that ability to create and use knowledge was a function of the degree of prior knowledge. We believed that the *interpretation role* of declarative knowledge and the *guidance role* of procedural knowledge [21] impacted team intelligence during an NPD project. In an NPD team setting, *declarative knowledge* denoted the teams' prior knowledge about a product category, experience, and familiarity with a product while *procedural knowledge* indicated the teams' prior knowledge and experience about customer needs and how to satisfy them in product design and its implementation [16]. Information on customers, competitors, and other parties is retained as declarative knowledge, which helps teams know what extra information is required, and how and where to find it to exploit new advances. In particular, NPD team members who are experts on processes and methods of product innovation are expected to generate, disseminate, utilize and respond to information in NPD processes. This is the foundation of the first set of hypotheses:

H1a. Team procedural knowledge is positively related to NPD team intelligence in projects.

H1b. Team declarative knowledge is positively related to NPD team intelligence in projects.

IT usage is another antecedent factor. From a KM context, Alavi and Leidner [6] noted that IT usage can create an infrastructure and environment that contributes to organizational KM by supporting, augmenting, and reinforcing knowledge processes at a deeper level.

During the NPD process, teams use IT tools, such as the Internet, Intranet, e-mails, teleconferencing, and FTP to communicate or store information. As IT tools help teams acquire and diffuse the information needed in the NPD process project teams can transform knowledge into value to their customers, and optimize the NPD process to fit the context of the project [25]. Therefore:

H2. IT usage during the NPD process is positively related to NPD team intelligence in projects.

In addition to knowledge base and IT effects, cross-functional diversity acts as another antecedent of team intelligence. It refers to the number of functional areas represented by people on the team. A greater number of functional areas are likely to improve input needed when making important product-related decisions. The necessity of team diversity occurs because cross-functional teams can generate and disseminate market and technology information, and thus increase responsiveness [24]. However, beyond some point, diversity can lead to increased decision complexity and confusion due to alternatives [28]. Sethi [27] suggests that teams simplify the heuristics or avoid in-depth processing of alternatives. Accordingly, we hypothesize that:

H3. Cross-functional diversity of project team will have an inverted-U relationship with NPD team intelligence in projects.

3.2. Consequences

Being creative or producing creative solutions is an essential result of organizational intelligence. Like this, individual knowledge has been stated to be important to individual creativity [30].

New Product Creativity is the degree to which a new product is novel and its introduction changes market thinking and practice. It denotes the extent to which the product generates ideas for other products, encourages fresh thinking, and challenges existing ideas for product category in the marketplace. We posit that team intelligence fosters new products that are more creative in the market place. Therefore:

H4. NPD team intelligence is positively related to NPC.

Finally, NPS was investigated as one of the consequences of team intelligence. NPS measures the market performance of a new product [7]. It is an important criterion for project managers and top management. The goal of forming an NPD project team and developing a new product is success of the product in the market place. Teams must detect and correct product related problems accurately and quickly, incorporate customer feedback and new technology, as appropriate, and facilitate flexible project structures. Then new products have more chance to become successful. Therefore, it is hypothesized that:

H5. NPD team intelligence is positively related to NPS.

4. Research design

4.1. Measures and sampling

To measure variables, we used or modified scale items based on the past research. Each construct was measured using multiple items and a 1–7 Likert scale (1 = strongly disagree to 7 = strongly agree). However, team size, which was selected as a control variable and measured as the number of persons in the team, and functional diversity questions were assessed with ratio scale, while NPC was measured with an itemized rating scale (1 = low to 7 = high). The questionnaire items are shown in Appendix A.

The initial sample consisted of 1250 U.S. manufacturing companies in the Project Management Institute (PMI). Product and project managers were used as key informants. The managers were chosen because, product/project managers,

- are likely to have a "big-picture view" of NPD projects,
- have a broader view of each member's behavior, and
- were expected to provide more reliable and objective data.

Since our study focused on the *NPD team as a unit of analysis*, product/project managers were likely to assess our variables more accurately. Also, the sample of respondents was similar to samples used in prior studies on innovation.

Each product/project manager in the PMI mailing list received an invitation to participate in our study (including a cover letter, the questionnaire, and stamped return envelope). Informants were asked to focus on their most recent product development projects (ones in the marketplace for a minimum of 12 months to ensure accurate assessment of the product performance). We received 207 usable responses (an effective response rate of 16.5%) representing a wide range of industries in the manufacturing sector, including: pharmaceuticals (28.5%), automotive (17%), electrical/electronics (17.9%), computers (13%), chemicals (8.7%), machinery/metal manufacturing (7.2%), and others (7.2%).

4.2. Measure validity and reliability

After data collection, the measures were statistically analyzed to assess their reliability, unidimensionality, discriminant validity, nomological validity, and convergent validity. An exploratory factor analysis was conducted, including 50 measured items of nine variables, using a principle component with a varimax rotation and an eigenvalue of one as the cut-off point (since team diversity and team sized was assessed with one item, those variables were not included in the analysis). A single factor was extracted for each multiple-item reflective scale. After performing this analysis, a subsequent confirmatory analysis was conducted to assess the scales.

A series of two-factor models was generated. In these, the factor correlations, considered individually, were restricted to unity and the fit of the restricted models was compared with that of the original model. The chi-square change ($\Delta\chi^2$) in all models, constrained and unconstrained, were significant ($p < 0.05$), confirming that the constructs demonstrated discriminant validity.

The measures were subjected to further CFA using EQS 5.7 [8]. This showed a good fit with an incremental fit index (IFI) of 0.97 and a comparative fit index (CFI) of 0.95 (with $\chi^2_{(36)} = 54$, RMSEA = 0.009). Further, we evaluated the validity of the measures used by linking eight constructs to new product success. During this analysis, we used "all items" for constructs. The model produced good fit indices ($\chi^2_{(335)} = 405$, CFI = 0.93, IFI = 0.94, RMSEA = 0.09), and each construct factor loading was large and significant, providing an adequate model fit.

NPD team intelligence was operationalized as a second-order construct with four dimensions. A second-order confirmatory

Table 1
Descriptive scales and construct correlations, and reliability estimates

	Variables	1	2	3	4	5	6	7	8	9	10	11
1	Information acquisition ability	–										
2	Information dissemination ability	0.15	–									
3	Information utilization ability	0.48***	0.54***	–								
4	Responsiveness	0.13	0.20**	0.19**	–							
5	Information technology usage	0.29***	0.48***	0.46***	0.18**	–						
6	Declarative knowledge	0.28***	0.31***	0.44***	0.15*	0.07	–					
7	Procedural knowledge	0.15*	0.20**	0.26***	0.21**	0.09	0.24***	–				
8	New product creativity	0.33***	0.20**	0.18**	0.45***	0.12	0.19**	0.21***	–			
9	New product success	0.06	0.12	0.16*	0.33***	0.06	0.19**	0.20***	0.03	–		
10	Functional diversity	0.13	0.08	0.04	0.11	0.09	0.16*	0.18*	0.19**	0.24***	–	
11	Team size	0.07	0.03	0.10	0.05	0.11	0.17*	0.22***	0.24***	0.20***	0.47***	–
	Mean	3.52	3.78	2.98	3.32	3.79	2.97	3.13	3.56	3.76	5.54	8.49
	Standard deviation	1.87	1.78	1.33	1.52	2.02	1.92	1.34	1.41	1.72	2.34	2.87
	Cronbach α	0.86	0.90	0.94	0.84	0.75	0.87	0.91	0.84	0.93	NA	NA

* $p < 0.1$.
** $p < 0.05$.
*** $p < 0.01$.

factor analysis of a model depicting information acquisition, dissemination and utilization ability and responsiveness was conducted. This model yielded acceptable fit indices ($\chi^2_{(57)} = 73$, CFI = 0.94, IFI = 0.94, RMSEA = 0.06). In addition, all first-order and second-order factor loadings were significant, thereby providing evidence that NPD team intelligence was a multifaceted construct. Hence, the second-order factor model demonstrated a composite NPD team intelligence in our study.

The reliability of the multiple-item, reflective measures, and the correlation among all 11 variables are reported in Table 1. The relatively low to moderate correlations provided further evidence of discriminant validity.

5. Analyses and results

Structural Equation Modeling (SEM) was performed using the maximum likelihood method (ML) to test the hypotheses. This assessed the integrity of the measures and evaluated the degree to which the observed relations among variables fitted the hypothesized network of causal relationships. The model included one second-order latent variable – *team intelligence* – consisting of *information processing capability*, which in turn had three sub-dimension, and *responsiveness capability*, which had one sub-dimension, and eight observed variables, declarative and procedural knowledge, functional diversity, functional diversity, IT usage, team size, NPS, and NPC: see Fig. 1. The conceptual model adequately fits the data: the normed fit index (NFI), and CFI exceeded 0.9. The RMSEA was 0.09, which was close to threshold level of 0.08.

The results indicated that a team's declarative ($\beta = 0.49$, $p < 0.05$) and procedural ($\beta = 0.67$, $p < 0.01$) knowledge were positively associated with the team intelligence, supporting H1a and H1b. The results also showed that IT usage had a positive association with the team intelligence ($\beta = 0.81$, $p < 0.01$)—supporting H2. Fig. 1 also shows that functional diversity was positively related to the team intelligence. However, the quadratic relationship between functional diversity with team intelligence ($\beta = 0.08$, $p > 0.1$) was not significant, therefore it did not support H3. We found that team intelligence was positively associated with NPC ($\beta = 0.72$, $p < 0.05$) and NPS ($\beta = 0.38$, $p < 0.05$), supporting H4 and H5.

Finally, Fig. 1 also showed that antecedent variables explained the 36% of variance ($R^2 = 0.36$) in team intelligence; team intelligence and control variable also explained 23% of variance ($R^2 = 0.23$) in NPC, and 12% of variance ($R^2 = 0.12$) in NPS.

6. Discussion

Our findings demonstrated that an efficiently managed NPD team knowledge has a positive influence on the success of a project. Our study also showed that team knowledge can be leveraged during the project. Especially, when the team knows about the product category; the competition, and the rules, and standards for developing and commercializing a new product. The knowledge base integrates team member's knowledge over time decreasing errors and duplication of effort. Also the knowledge base helps the project team to recombine successful experience leading to a superior solution quickly.

Interestingly, we did not find a quadratic relation between functional diversity and team knowledge. Rather we found that the higher the number of functional areas represented on the team, the higher the team's ability to acquire, process, and utilize knowledge and become flexible to change in customer needs and wants, etc.

7. Managerial implications

Managers should enhance team intelligence to facilitate the success of an NPD project in the manufacturing industry by implementing a well-defined IT system for effective team communication and information exchange and installing e-mail systems, team messaging boards, electronic newsletters, Web page, and verbal electronic communication tools for the enhancement of the team's knowledge base.

Management should form and organize the project team based on the team members' previous experiences and skills and bring diversification to the team with a variety of people from the different functional departments.

Management should set up customer councils, panels, and groups to facilitate interactions between project team members and customers and create knowledge bases from previous projects and studies to aid in re-examining past learning.

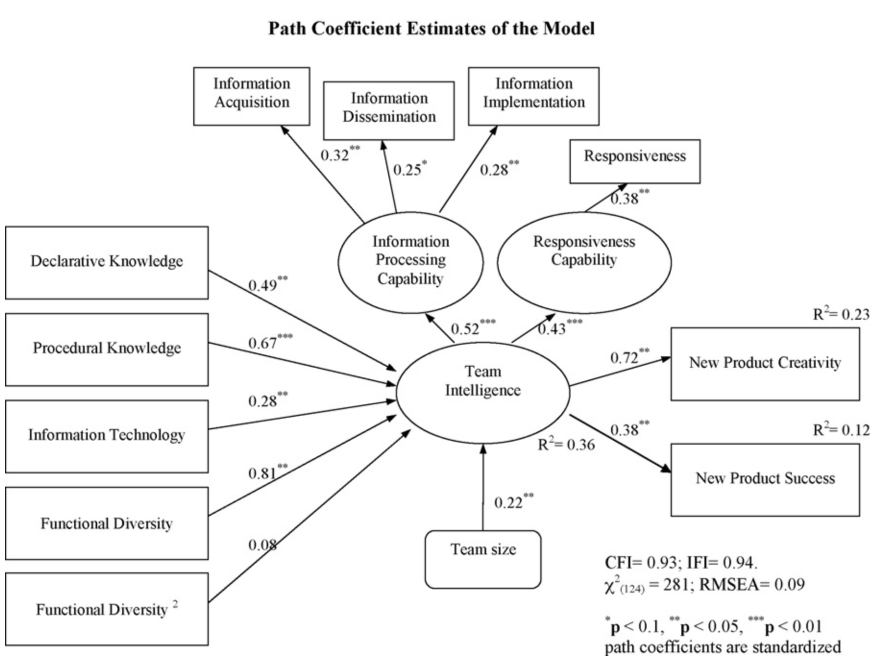

Fig. 1. Path coefficient estimates of the model.

8. Limitations

There are some methodological limitations in our study. It was prone to common method bias since the same respondents answered the dependent variable and the independent variable in a cross-sectional manner. We checked this potential problem with a Harman one-factor test [26]. An unrotated factor analysis of nine focal variables resulted in a nine-factor solution, as expected, this accounted for 86.4% of the total variance; and factor 1 accounted for 19.2% of the variance. Because a single factor did not emerge and factor 1 did not explain most of the variance, common method bias is unlikely to be of concern in our data.

Utilizing a cross-sectional design with questionnaires was also a limitation of our study. A longitudinal research method should probably been used to obtain more objective results about the flow of knowledge. Also, using objective measures, such as archival data, have produced results that were more objective.

Although the heterogeneity of the sample, including a myriad of industries within the manufacturing category, was a condition for external validity and generality, sector or industry level studies may have been useful in validating the results. For instance the model could have been tested in software industry. Since there are differences between software development and NPD project teams, the conclusions of the model may be different. Further, the results could be different due to the maturity and competitive dynamics of the various industries.

9. Conclusion

Literature on organizational and team intelligence demonstrated that information processing and responsiveness capabilities were important for the performance of organizations and teams. However, there is still a lack of operationalization, and empirical test of antecedents and consequences of intelligence in NPD teams. Our study measured team intelligence with its antecedents and consequences. We showed empirically that team intelligence impacted NPC and NPS, and the team's knowledge base (declarative and procedural knowledge), functional diversification, and IT usage influenced the project team intelligence.

Appendix A. Measures

We used a Likert scale (0 = strongly disagree to 7 = strongly agree):

- *Information acquisition ability* (modified from Moorman [22])—during this project, the team had the *ability* of continuously:
  - Collecting information from customers.
  - Collecting information about competitors' activities.
  - Collecting information about relevant publics other than customers and competitors.
  - Re-examining the value of information collected in previous studies.
  - Collecting information from external experts, such as consultants.
- *Information dissemination ability* (modified from Moorman [22])—during this project, the team was *able to* have:
  - Formal information links established among all parties involved in the project.
  - Informal networks that ensured all team members generally had the information they needed.
- During this project, the team members were able to:
  - Educate each other during the project.
  - Be trained in new tasks relating to this project.
- *Information utilization ability* (modified from Moorman [22])—during this project, the team was able to:
  - Summarize information, reducing its complexity.
  - Organize information in meaningful ways.
  - Process information in meaningful ways.
  - Rely heavily upon information to make decisions relating to the project.
  - Use information to solve specific problems encountered in the project.
  - Provide information to effectively implement the project.
- *Responsiveness* (adapted from Akgün et al. [4])—during this project, the team had the *ability to*:
  - Transfer customer needs to product design specs.
  - Generate different market and technology scenarios.
  - Take corrective action immediately, when they found out the changes in customer preferences, wants, and needs.
  - Respond to significant changes in our business environment.
- *Information technology usage* (derived from Wakefield [32] and Agius and Angelides [2])—during this project, the team relied on the following tools to communicate with team members:
  - E-mail to fellow team members.
  - Team messaging boards or team discussion forums.
  - Shared electronic files.
  - Electronic newsletters that covered project information.
  - A web page dedicated to this project.
  - Teleconferencing.
  - Attaching audio/video files to electronic documents.
- *Declarative Knowledge* (adapted from Moorman and Miner [21])—*prior to the project*, the team had a great deal of knowledge about ...
  - This product category.
  - Competitors and their products.
  - Procedures, rules, and standards on developing a new product.
  - Procedures, rules, and standards on commercializing a new product.

  Procedural Knowledge (adapted from Moorman and Miner [21])—*prior to the project*, the team had a great deal of ...
  - Experience on how to develop a new product.
  - Skills on how to develop a new product.
  - Experience on how to commercialize a new product.
  - Skills on how to commercialize a new product.
- *New Product Creativity* (adapted from Moorman [22])—this product:
  - challenged existing ideas for this category,
  - offered new ideas to the category,
  - was creative,
  - was interesting,
  - was very novel for this category,
  - spawned ideas for other products, and
  - encouraged fresh thinking.
- *New Product Success* (adapted from Cooper and Kleinschmidt [10])—this product:
  - Met or exceeded volume expectations.
  - Met or exceeded sales dollar expectations.
  - Met or exceeded the first year number expected to be produced and commercialized.
  - Overall, met or exceeded sales expectations
  - Met or exceeded profit expectations.
  - Met or exceeded return on investment (ROI) expectations.
  - Met or exceeded overall senior management's expectations.
  - Met or exceeded market share expectations.
  - Met or exceeded customer expectations.

- *Functional diversity* (adapted from Sethi [27]):

The number of functional areas (departments) represented on the team whose members were fully involved in the project rather than being ad hoc specialists or consultants who were engaged only for a limited time.

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Ali E. Akgün is an associate professor of science and technology studies in the School of Business Administration at Gebze Institute of Technology, Turkey. He received his PhD in technology management from Stevens Institute of Technology and his MS in engineering management from Drexel University. His research has been published in the Human Relations, Information & Management, Journal of Engineering and Technology Management (JET-M), Journal of Product Innovation Management, IEEE Transaction in Engineering Management, Industrial Marketing Management, European Journal of Information Systems, Research Technology Management among other journals. His research areas are new product/technology development, organizational learning, and cognitive and social psychology.

Mumin Dayan is an assistant professor of marketing in the College of Business Administration at University of Sharjah. He received his MBA degree in marketing from Drexel University and PhD in marketing from the Fox School of Business, Temple University. His research interests include innovation and knowledge management in new product development teams and small businesses.

C. Anthony Di Benedetto is professor of marketing and senior washburn research fellow at the Fox School of Business and Management, Temple University. He received a BSc (chemistry), an MBA, and a PhD in marketing and management science from McGill University. He serves as editor in chief of the Journal of Product Innovation Management. His research interests include international marketing strategy, new product launch decisions, radical innovation, and supplier-manufacturer-distributor cooperation. His research articles have appeared in J. of Product Innovation Management, Management Science, Strategic Management Journal, J. of International Business Studies, and elsewhere.